Computer Architecture: Neuromorphic Processing Unit

ธัชพงศ์ กตัญญูกุล

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MIT Technology Review



Agricultural Drones
Relatively cheap drones with advanced sensors and imaging capabilities are giving farmers new ways to increase yields and reduce crop damage.



Ultraprivate Smartphones
New models built with security and privacy in mind reflect the Zeitgeist of the Snowden era.



Brain Mapping
A new map, a decade in the works, shows structures of the brain in far greater detail than ever before, providing neuroscientists with a guide to its immense complexity.



Neuromorphic Chips
Microprocessors configured more like brains than traditional chips could soon make computers far more astute about what's going on around them.



Genome Editing
The ability to create primates with intentional mutations could provide powerful new ways to study complex and genetically baffling brain disorders.



Microscale 3-D Printing
Inks made from different types of materials, precisely applied, are greatly expanding the kinds of things that can be printed.



Mobile Collaboration
The smartphone era is finally getting the productivity software it needs.

10 Breakthrough Technologies 2014

Introduction
Agricultural Drones
Ultraprivate Smartphones
Brain Mapping
Neuromorphic Chips
Genome Editing
Microscale 3-D Printing
Mobile Collaboration
Oculus Rift
Agile Robots
Smart Wind and Solar Power
Archive of Past Lists

Neuromorphic Processing Unit

Neuromorphic Chips:

- Traditional chips are reaching fundamental performance limits
- an alternative design for computer chips that will enhance artificial intelligence
- Key Players: IBM, Qualcomm, HRL Laboratories, Human Brain Project

	what they do well	what they're good for
Traditional chips (von Neumann architecture)	Reliably make precise calculations	Anything that can be reduced to a numerical problem, although more complex problems require substantial amounts of power
Neuromorphic chips	Detect and predict patterns in complex data, using relatively little electricity	Applications that are rich in visual or auditory data and that require a machine to adjust its behavior as it interacts with the world

Artificial Brain: IBM's TrueNorth

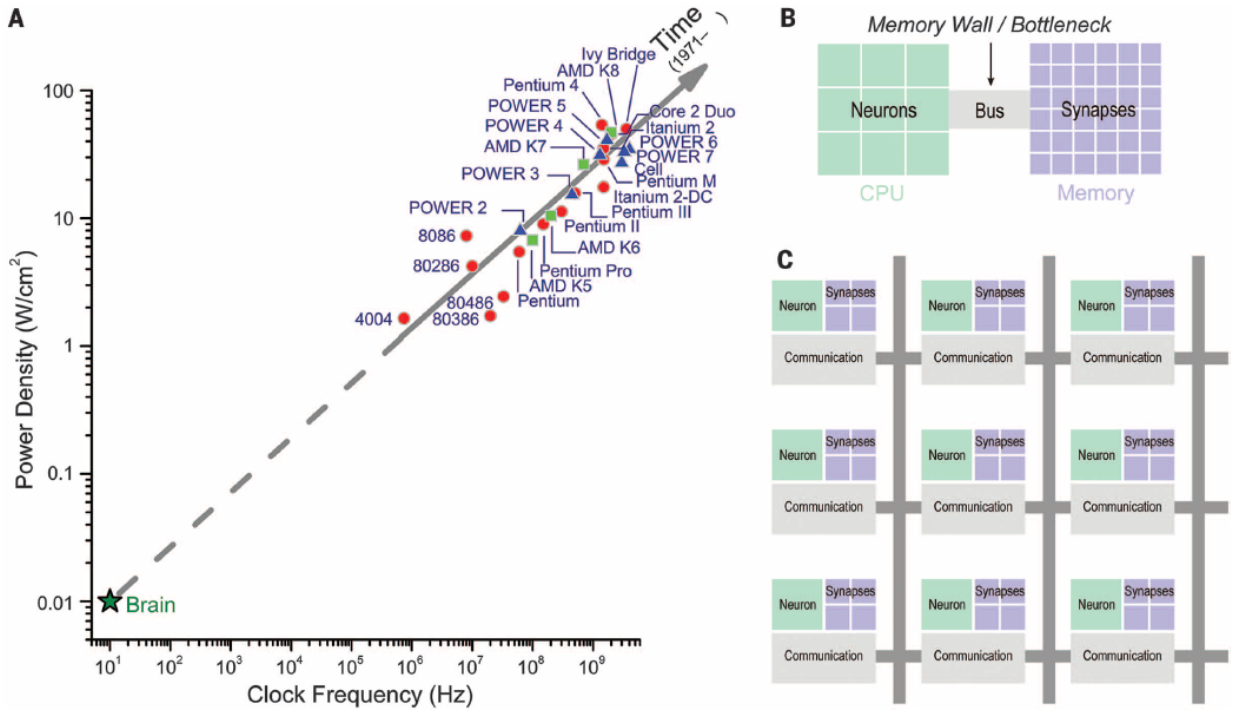
Merolla, et. al., A million spiking-neuron integrated circuit with a scalable communication network and interface, SCIENCE 345(6197), 8 Aug 2014.

"Inspired by the brain's structure, we have developed an efficient, scalable, and flexible non-von Neumann architecture that leverage contemporary silicon technology.
To demonstrate, we built a 5.4-billion-transistor chip with 4096 neurosynaptic cores interconnected via an intrachip network that integrates 1 million programmable spiking neurons and 256 million configurable synapses.

... With 400-pixel-by-240-pixel video input at 30 frames per second, the chip consumes 63 milliwatts."

TrueNorth: Motivation

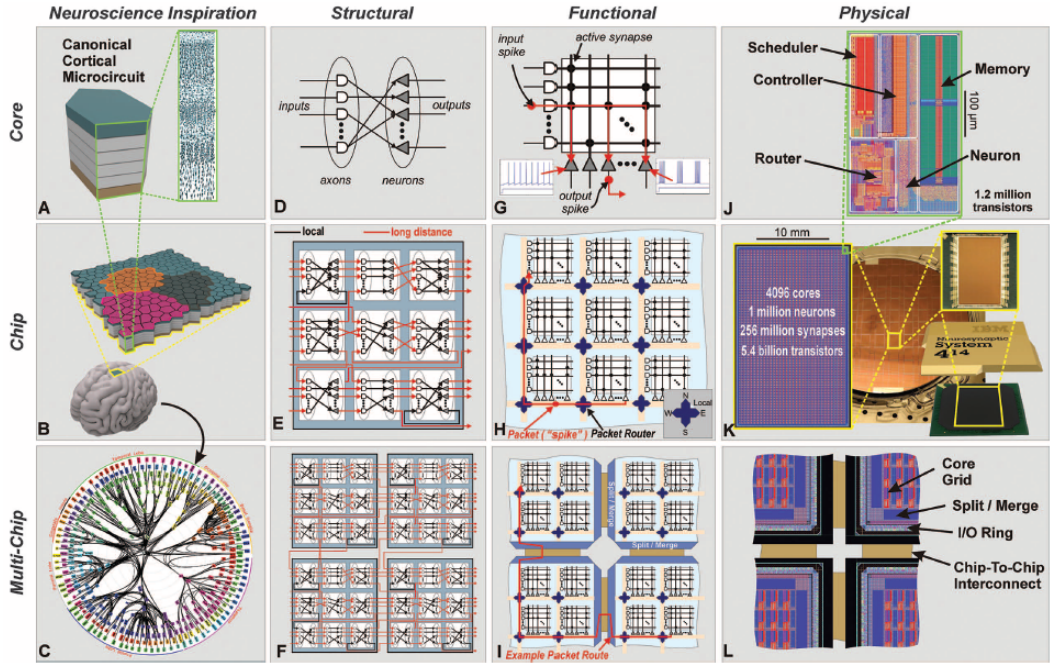
"The von Neumann architecture is fundamentally inefficient and nonscalable for representing massively interconnected neural networks ..."



All pictures and materials are from Merolla, et. al. 2014

TrueNorth: Idea

"... the basic building block is a core, a self-contained neural network with 256 input lines (axons) and 256 outputs (neurons) connected via 256-by-256 directed, programmable synaptic connection."

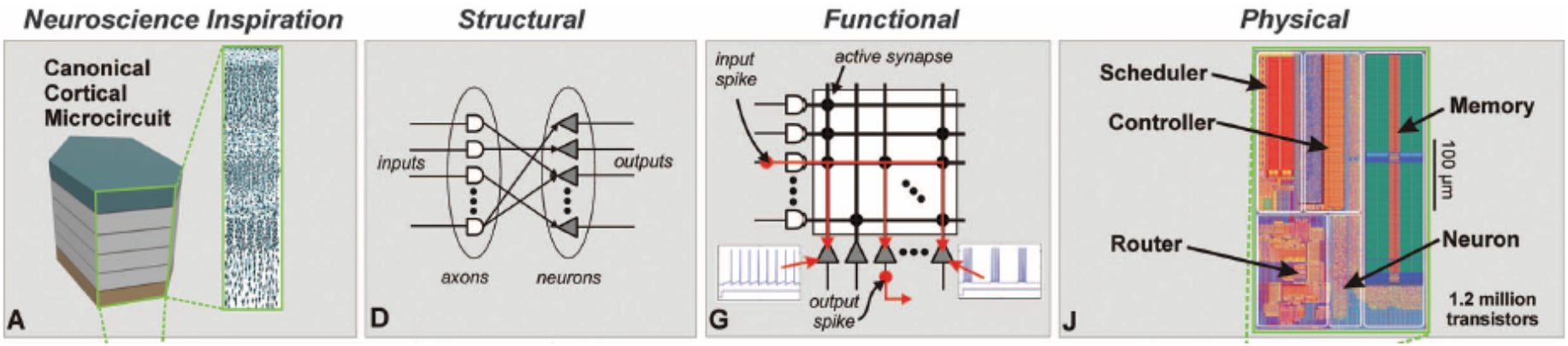


All pictures and materials are from Merolla, et. al. 2014

TrueNorth: Rationales

Efficiency

- (i) neurons form clusters that draw their inputs from a similar pool of axon (Fig. 2A) allowing for memory computation colocalization;
- (ii) only spike events, which are sparse in time, are communicated between cores via the long-distance communication network; and
- (iii) active power is proportional to firing activity.

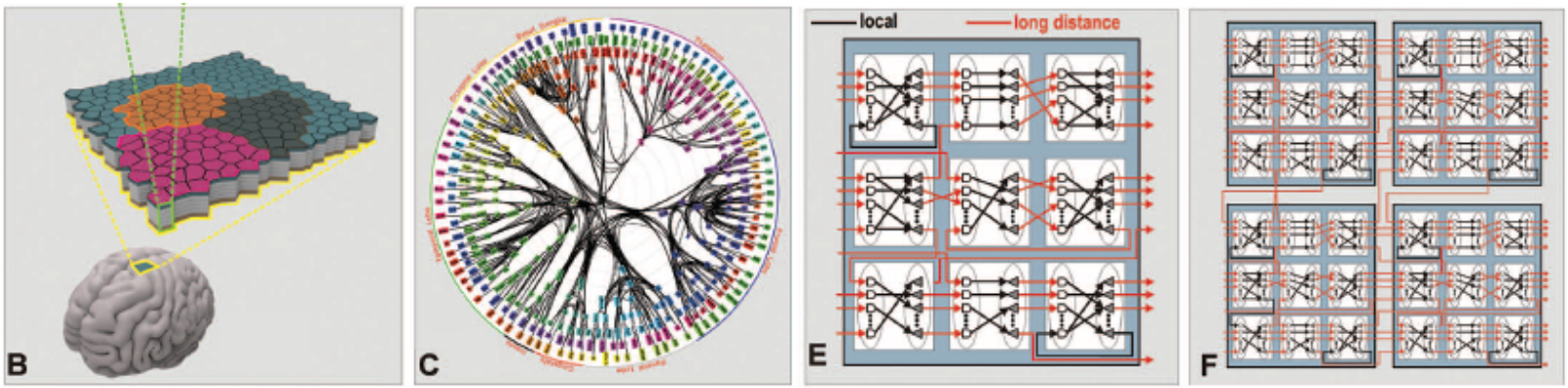


All pictures and materials are from Merolla, et. al. 2014

TrueNorth: Rationales

Scalability

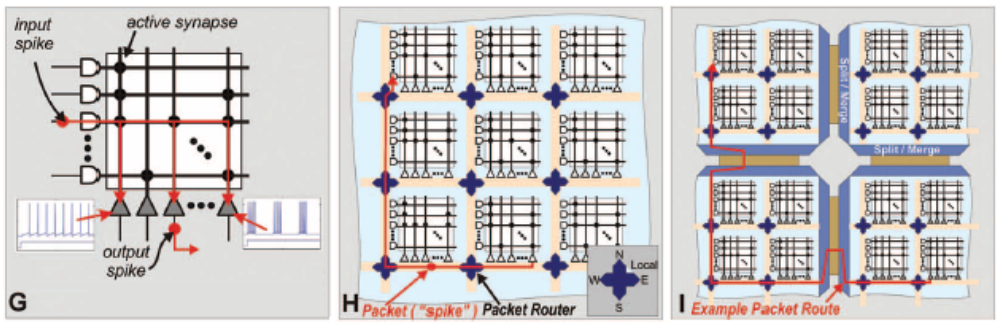
- (i) cores on chip, as well as chips themselves, can be tiled in two dimensions similar to the mammalian neocortex (Fig. 2, B and C);
- (ii) each spike event addresses a pool of neurons on a target core, reducing the number of long-range spike events thus mitigating a critical bottleneck; and
- (iii) occasional defects at the core and chip level do not disrupt system usability.



All pictures and materials are from Merolla, et. al. 2014

TrueNorth: Flexibility

- (i) each neuron is individually configurable, and the neuron model supports a wide variety of computational functions and biologically relevant spiking behaviors;
- (ii) each synapse can be turned on or off individually, and postsynaptic efficacy can be assigned relative strengths;
- (iii) each neuron-axon connection is programmable along with its axonal delay; and
- (iv) the neurons and synapses can exhibit programmed stochastic behavior via a pseudo-random number generator (one per core).
- thus supports feed-forward, recurrent, lateral connections, etc.

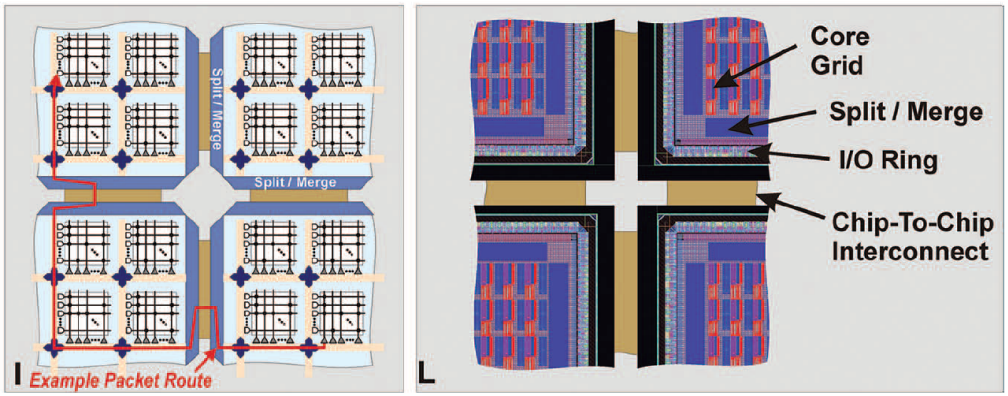


All pictures and materials are from Merolla, et. al. 2014

TrueNorth: Architecture

Hardware/Software Interface

- TrueNorth takes spike events as its input
- and gives spike events as its output

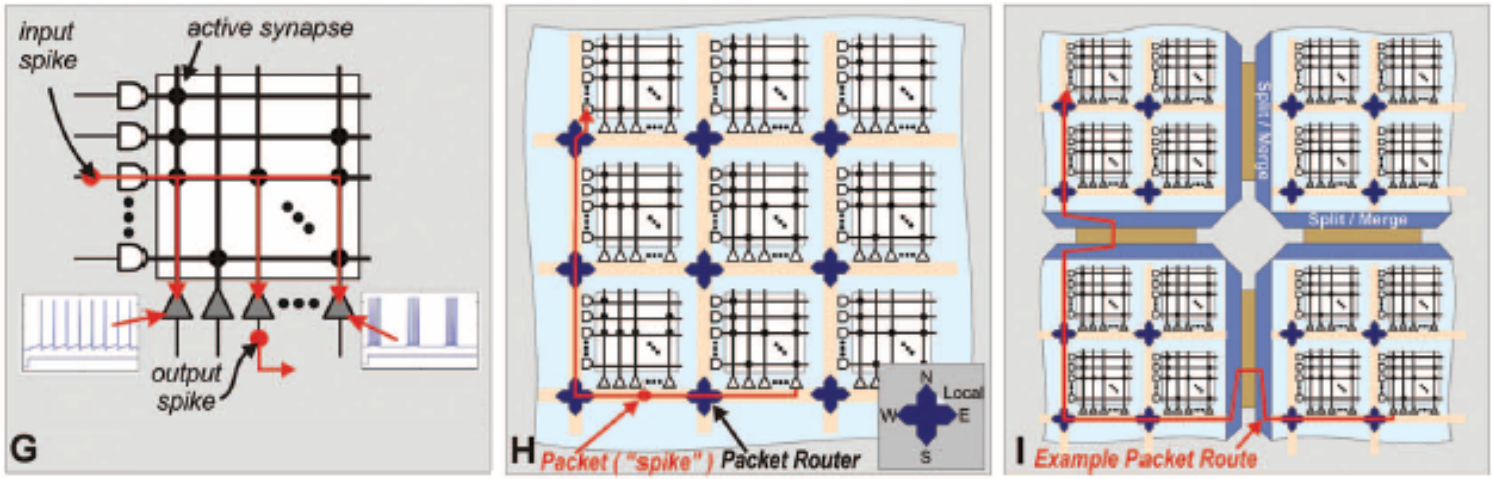


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TrueNorth: Architecture

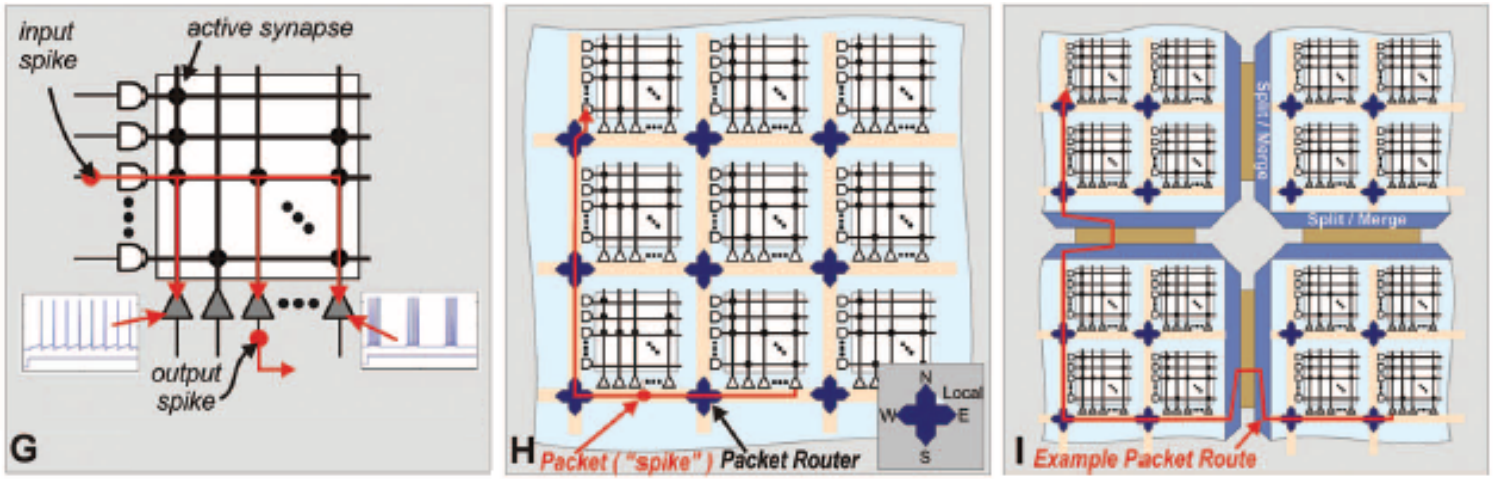
Programming

- addressable axons ($\equiv$ configurable inputs)
- configurable synaptic crossbar array ($\equiv$ configurable input-output relation)
- programmable neurons ($\equiv$ configurable output function)
- program: specifying connections and neuron functions so that we get correct output spike events corresponding to given input spike events



All pictures and materials are from Merolla, et. al. 2014

TrueNorth: How It Works



SYNAPSE & LEAK INTEGRATION

$$V_j(t) = V_j(t-1) - \lambda_j + \sum_{i=0}^{255} A_i(t) \times w_{i,j} \times S_j^{G_i}$$

THRESHOLD, FIRE, RESET

if $V_j(t) \geq \alpha_j$

Spike

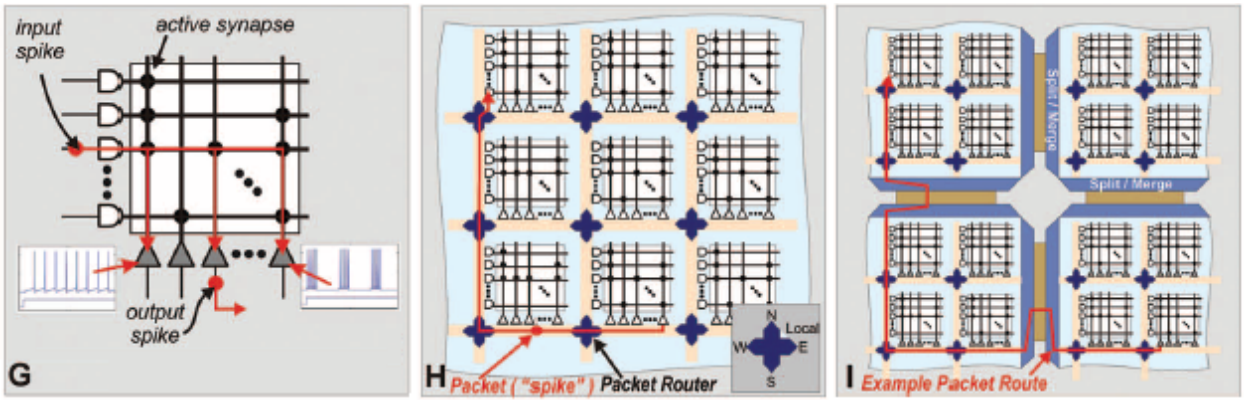
$V_j(t) = R_j$

endif

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TrueNorth: How It Works

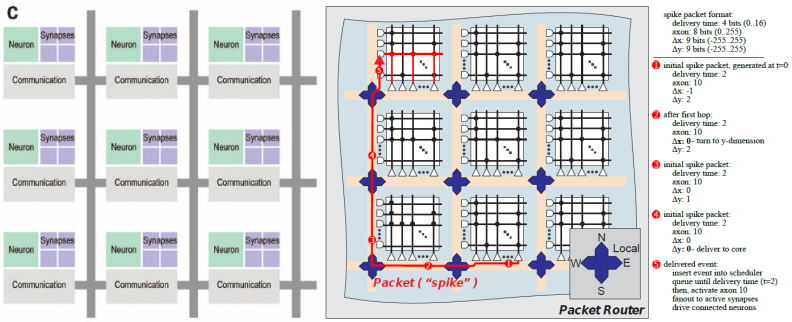
- Axons are activated by input spike events, which are generated by neurons anywhere in the system and delivered after desired axonal delay of between 1 and 15 time steps.
- spike events are carried between cores by time-multiplexed wires that interconnect a two-dimensional mesh network of routers, each with five ports (north, south, east, west, and local).
- The routers form the backbone of a two-dimensional mesh network interconnecting a 64-by-64 core array (Fig. 2H).



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TrueNorth: How It Works

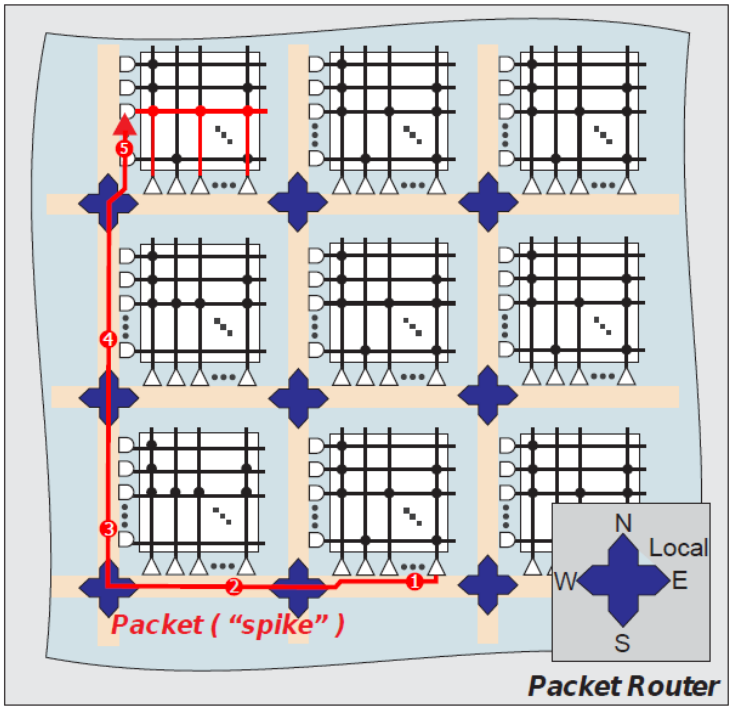
- When a neuron spikes, it looks up in local memory an axonal delay (4 bits) and the destination address (8-bit absolute address for the target axon and two 9-bit relative addresses representing core hops in each dimension to the target core).
- This 30-bit connection information is encoded into a packet that is injected into the mesh.
- The packet is handed from core to core–first in the x dimension then in the y dimension (deadlock-free dimension-order routing) until it arrives at its target core before fanning out via the crossbar (Fig. S2).



All pictures and materials are from Merolla, et. al. 2014

TrueNorth: How It Works

- To implement feedback connections within a core, where a neuron connects to an axon on the same core, the packet is delivered by using the router's local channel. (Efficient: it never leaves the core.)



spike packet format:
delivery time: 4 bits (0..16)
axon: 8 bits (0..255)
Δx: 9 bits (-255..255)
Δy: 9 bits (-255..255)

❶ initial spike packet, generated at t=0:
delivery time: 2
axon: 10
Δx: -1
Δy: 2

❷ after first hop:
delivery time: 2
axon: 10
Δx: 0- turn to y-dimension
Δy: 2

❸ initial spike packet:
delivery time: 2
axon: 10
Δx: 0
Δy: 1

❹ initial spike packet:
delivery time: 2
axon: 10
Δx: 0
Δy: 0- deliver to core

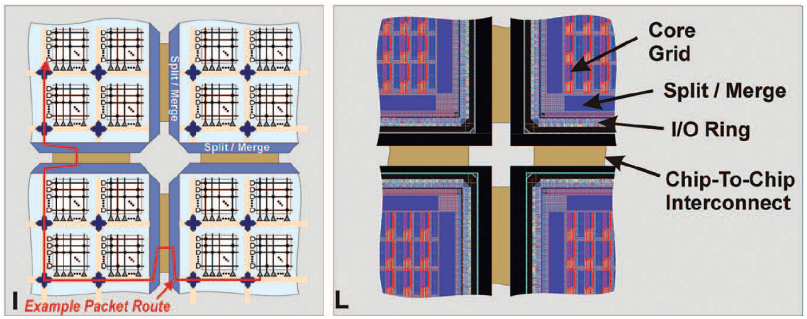
❺ delivered event:
insert event into scheduler
queue until delivery time (t=2)
then, activate axon 10
fanout to active synapses
drive connected neurons

All pictures and materials are from Merolla, et. al. 2014

Scaling TrueNorth

To scale the 2-D mesh across chip boundaries where the number of interchip connections is limited,

- TrueNorth uses a merge-split structure at the four-edges of the mesh to serialize exiting spikes and deserialize entering spikes (Fig. 2I).
- Spikes leaving the mesh are tagged with their row (for spikes traveling east-west) or column (traveling north-south) before being merged onto a shared link that exits the chip.
- Spikes entering the chip from a shared link are split to the appropriate row or column by using the tagged information.

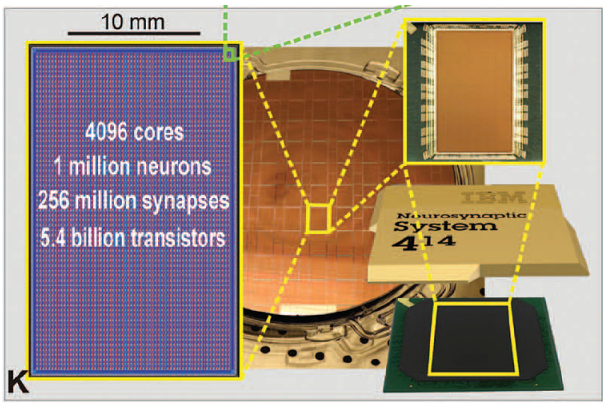


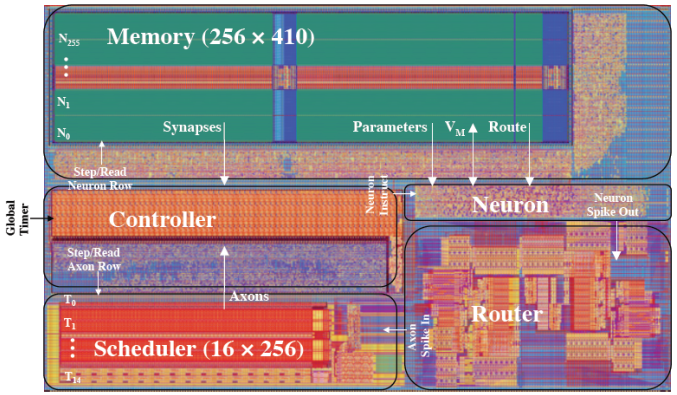
All pictures and materials are from Merolla, et. al. 2014

TrueNorth: Implementation

TrueNorth

- fully functional digital chip: 5.4 billion transistors on 4.3-cm^2 area in Samsung's 28-nm process technology
- 4096 cores a chip: each core has 256 neurons, 256 axons $\equiv\sim$ 1M neurons a chip
- $\sim$ 428 million bits of on-chip memory
  104,448 bits of local memory on each core: $65,536$ bits for synaptic states, $31,232$ bits for neuron states and parameters, $6,656$ bits for destination addresses (each neuron can link up to 26 axons), and $1,024$ bits for axonal delay.





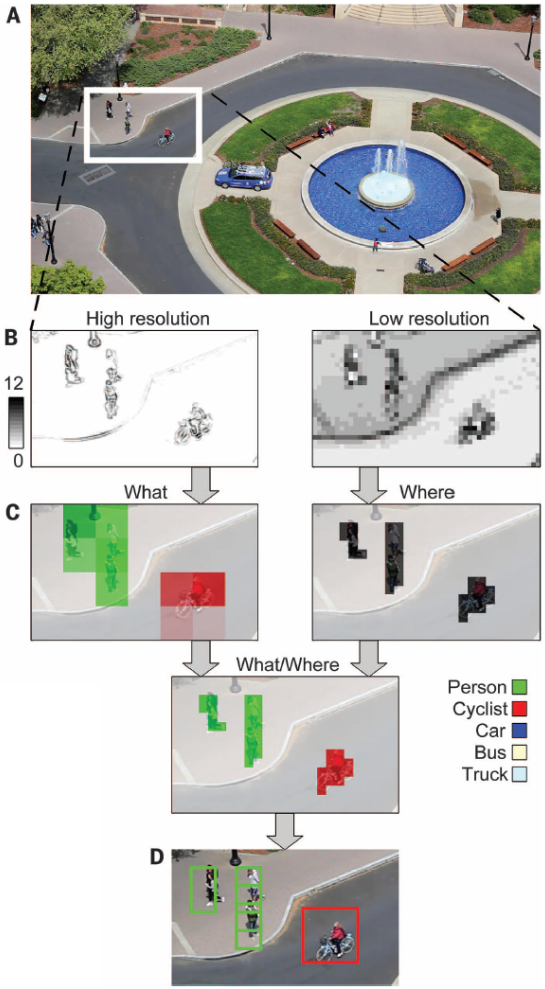
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TrueNorth: Evaluation

- TrueNorth's power density is 20 mW/cm^2 c.f. typical CPU $\sim$ 50 to 100 W/cm^2
- Merolla et. al. uses their software ecosystem to map many well-known algorithms to TrueNorth via offline learning.
  E.g., convolutional networks, liquid state machine, restricted Boltzmann machines, hidden Markov models, support vector machines, optical flow, and multimodal classification can be run on TrueNorth without modification.
- Merolla et. al. tested TrueNorth on multiobject detection and classification:
  - (i) detect people, bicyclists, cars, trucks, and buses
  - (ii) consumes 63 mW to process 400 $\times$ 240-pixel images at 30-frame-per-second of 3-color video

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TrueNorth: Evaluation

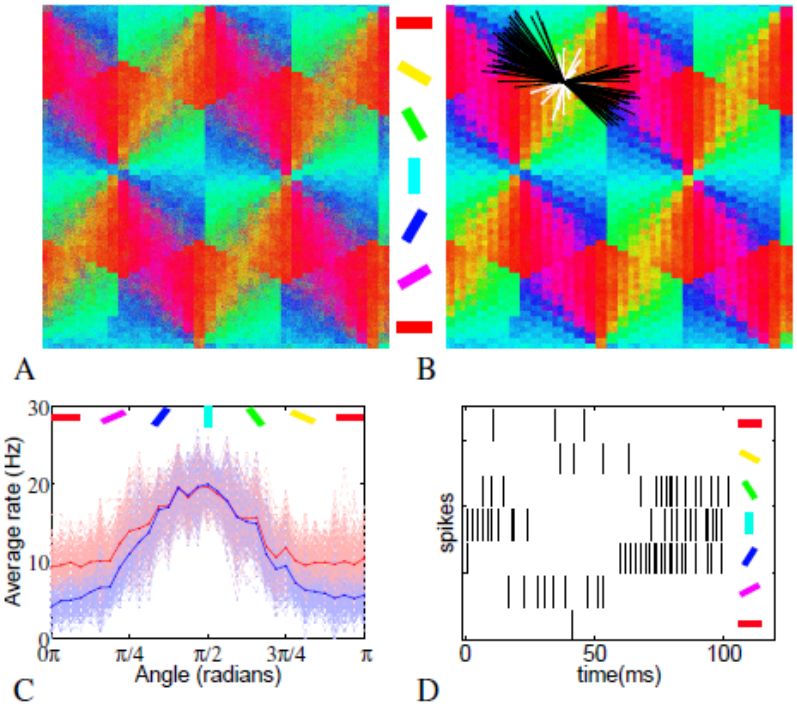


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- Neovision2 Tower data set
- identify the labels and locations of objects among five classes: person, cyclist, car, bus, truck.
- The region is transduced from pixels into spike events to create two parallel channels:
  - a high resolution channel for labeling objects ($\sim$ dorsal pathway in visual cortex)
  - a low resolution channel for locating salient objects ($\sim$ ventral pathway in vsual cortex)

TrueNorth: Evaluation

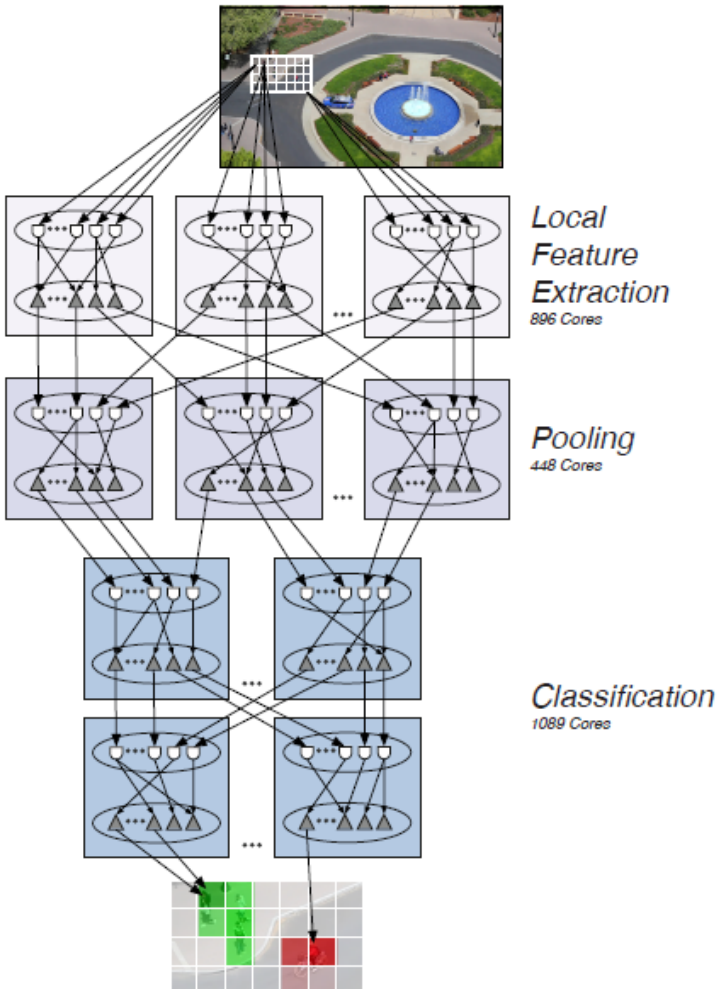
- The pixels are converted into spike events to interface with TrueNorth



All pictures and materials are from Merolla, et. al. 2014

- (A) Feedforward orientation preference map of neuron selectivity, across TrueNorth's one million neurons, organized as a $1,024 \times 1,024$ grid.
- (B) Orientation map with added lateral connections, where similar orientations excite each other (white) and dissimilar ones inhibit (black), sharpens selectivity on average by 86% while increasing power by 1%.
- (C) Averaged tuning curves for neurons selective for vertical orientation ($\pi/2$) with lateral connections (blue) and without (red). Adding lateral connections increases selectivity by reinforcing the prefered orientation while suppressing non-preferred ones.
- (D) Example spike responses from a vertically-selective neuron for different orientation patterns.

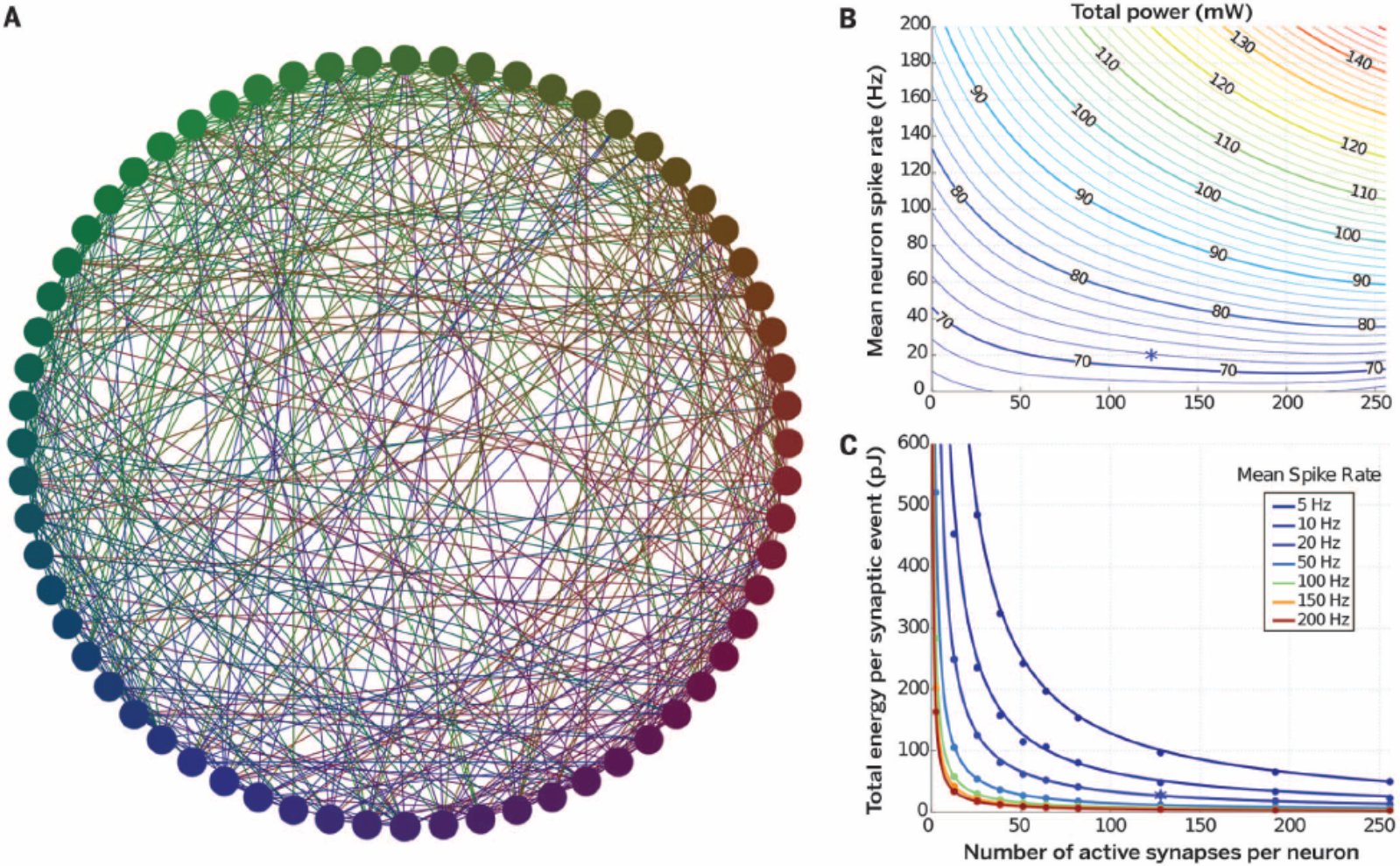
TrueNorth: Evaluation



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- Classification system network is the largest block in the object detection and classification application.
- It consists of three layers of integrate-and-fire neurons with their parameters and connections configured to perform local feature extraction, pooling and classification.

TrueNorth: Results



All pictures and materials are from Merolla, et. al. 2014

TrueNorth: Interpreting Results

- At the operating point where neurons fire on average at 20 Hz and have 128 active synapses, the total measured power was 72 mW (at 0.775 V operating voltage)
  - $\equiv$ 26 pJ per synaptic event
  - c.f. running the same algorithm on a modern general-purpose microprocessor, TrueNorth consumes 176,000 times less energy per event.
  - c.f. a state-of-the-art multiprocessor neuromorphic approach (Stromatias et al. 2013), TrueNorth consumes 769 times less energy per event.
- Computation in TrueNorth is measured by using synaptic operations per second (SOPs)

  c.f. floating-point operations per second (FLOPs)
- TrueNorth: 46 billion SOPs per watt (typical) and 400 billion SOPs per watt (high spike rates and high number of active synapses)

  c.f. "today's most energy-efficient supercomputer achieves 4.5 billion FLOPs per watt."

TrueNorth: Results

		Probabilistic Network	Visual Filter	Object Detection & Recognition
TrueNorth	Time(s)	10	10	10
	Power(W)	0.072	0.060	0.063
	Energy(J)	0.720	0.600	0.630
	Energy Ratio (Energy / TrueNorth Energy)	1.000	1.000	1.000
CPU system	Time(s)	1,290	2,012	1,025
	ΔPower(W)	98.3	90.7	80.1
	Energy(J)	127,000	182,000	82,000
	Energy×TN (Energy / TrueNorth Energy)	176,000	304,000	130,000

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Wisdom Quote